

Geometry

Unit 2

Lesson 2 Homework

Name **Answer Key**

Date _____

Period _____

In questions 1-5, determine the type of reasoning for each statement.

1. In order to participate in school sports, students must have a minimum GPA of a 2.0. Jim has a GPA of 2.75, therefore he can participate in school sports.

Deductive Reasoning

2. Every Tuesday that Yolanda's family has visited a specific local restaurant, the kids' meals have been free. When Yolanda's family visits that restaurant next Tuesday, the kids' meals will be free.

Inductive Reasoning

3. A new movie director has grossed over 250 million for each of his first three movies. When his fourth movie is released, it will gross over 250 million.

Inductive Reasoning

4. Any given number multiplied by one equals itself. If 5 is multiplied by one, it will equal itself.

Deductive Reasoning

5. The air conditioner is set to keep a particular home at 72°F and runs off the home's electricity. During a power outage, the home will not stay at 72°F.

Deductive Reasoning

For each statement in 6-10, write a counterexample that disproves the statement.

6. Mary enters her home and her clothes are wet. Therefore, it is raining outside.

Answers may vary.

The sprinklers are running outside and Mary walked in their path.

7. If a number is multiplied to another number, it will result in a larger number.

Answers may vary. $5 \times 0 = 0$

8. If a figure has four sides and all sides have the same length, it is a square.

A rhombus has four sides with the same length and is not a square.

9. If a number is prime, then it is an odd number.

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10. Yvette lives in Jacksonville. Therefore, she lives in Florida.

Answers may vary. Yvette lives in Jacksonville, Alabama.